A DIRECT SUMMAND OF A FINITE LOCALLY FREE MODULE IS FINITE LOCALLY FREE WHEN THE LOCAL RINGS ARE LOCAL

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The structure of finitely generated projective modules over a local ring as finite free modules. Invoked inside the counted proof at source lines 2404-2406 as a cross-chapter pointer. Identifies the stalk of a summand of a finite free module over a local ring as free of a rank depending on the point.
- The criterion for a module of finite presentation whose stalk at a point is free to be free of that rank on a neighbourhood. Invoked inside the counted proof at source lines 2407-2408 as a pointer to a lemma of this chapter. Spreads the local freeness of the stalk to an open neighbourhood of the point.

Source reference. Original reference algebra-lemma-finite-projective: sources/stacks-modules/pinned-original-source.tex, source lines 2406–2406. Navigation pointer only: the counted text refers to the Stacks item algebra-lemma-finite-projective, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-finite-presentation-stalk-free: sources/stacks-modules/pinned-original-source.tex, source lines 2407–2407. Navigation pointer only: the counted text refers to the Stacks item lemma-finite-presentation-stalk-free, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

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Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. If all stalks $\mathcal{O}_{X,x}$ are local rings, then any direct summand of a finite locally free $\mathcal{O}_X$-module is finite locally free.*

Proof. Assume $\mathcal{F}$ is a direct summand of the finite locally free $\mathcal{O}_X$-module $\mathcal{H}$. Let $x \in X$ be a point. Then $\mathcal{H}_x$ is a finite free $\mathcal{O}_{X,x}$-module. Because $\mathcal{O}_{X,x}$ is local, we see that $\mathcal{F}_x \cong \mathcal{O}_{X,x}^{\oplus r}$ for some r, see Algebra, Lemma algebra-lemma-finite-projective. By Lemma lemma-finite-presentation-stalk-free we see that $\mathcal{F}$ is free of rank r in an open neighbourhood of x. (Note that $\mathcal{F}$ is of finite presentation as a summand of $\mathcal{H}$.) $\square$